

Cannabis Access and College Persistence: Evidence from Washington State’s Dispensary Lottery

Anand Mathew*

April 28, 2025

Abstract

This paper examines how proximity to recreational marijuana dispensaries affects postsecondary educational outcomes in Washington State. Leveraging the quasi-random placement of dispensaries via a state-run license lottery and a staggered difference-in-differences framework, I estimate the causal impact of dispensary access on college enrollment and retention rates. Using campus-level administrative data from the Integrated Postsecondary Education Data System (IPEDS) and detailed dispensary location data, I find that institutions located within a 10-minute driving radius of a dispensary experienced a statistically significant 7.9 percentage point decline in student retention rates. In contrast, there is no evidence that dispensary proximity affects college enrollment. Robustness checks across alternative distance thresholds confirm that the retention effect is stable and significant for thresholds between 10 and 15 minutes, while enrollment results remain null. These findings suggest that while local cannabis availability may not deter students from enrolling, it can negatively affect persistence among those already enrolled. The study contributes to emerging literature on the localized educational consequences of marijuana legalization and highlights the importance of spatial exposure in shaping policy impacts.

1 Introduction

Cannabis policy in the United States has undergone dramatic changes in the past decade. Once uniformly prohibited, recreational marijuana use is now legal in nearly half of U.S. states. Washington State, alongside Colorado, became the first states to legalize recreational cannabis in 2012 through a voter-approved initiative (Mosher and Akins, 2020). With no prior precedent, many predicted severe negative consequences, sparking intense debates about the potential effects on various social, economic, and public health outcomes. A growing body of research has sought to investigate the effects of recreational marijuana legalization (RML) on crime, substance use, local economic activities, and education (Anderson and Rees, 2014; Hansen et al., 2017; Tyndall, 2021; Nicholas and Maclean, 2019; Dong,

*Department of Economics, University of Florida.

2022). This paper contributes to the literature by estimating the short-term causal impacts of recreational marijuana dispensaries on college educational outcomes, specifically student retention and enrollment, in Washington State.

The passage of Initiative 502 (I-502), which legalized the possession and sale of marijuana for adults aged 21 and over, was met with significant opposition. Critics raised concerns about the potential impact on driving under the influence and the risks it posed to minors. While some argue that marijuana legalization could reduce school dropout rates by decreasing youth involvement in the illicit drug market (Becker and Murphy, 2013), others have cautioned that it may lead to negative consequences, such as increased substance use and lower academic performance (Chan et al., 2024).

This study employs a staggered difference-in-differences (DiD) approach, taking advantage of the timing of dispensary openings in Washington State. An institution is designated as 'treated' if it is within 10 minutes of driving time to an open dispensary. This treatment status changes over time as dispensaries open, allowing for a comparison between 'treated' and 'control' universities, where the latter are located farther from dispensaries. By using this approach, I am able to examine how the proximity of dispensaries affect student outcomes across different institutions.

I find that the retention rate at universities with a marijuana dispensary within a 10-minute driving distance decreased by 7.9% on average, while no significant effect was observed on overall college enrollment rates.

By focusing on Washington's unique lottery system for granting dispensary licenses, this study contributes to the ongoing policy debate regarding the unintended consequences of marijuana legalization. It provides new insights into the local impact of dispensaries on higher education outcomes and informs ongoing discussions on balancing public health objectives with educational priorities.

The remainder of the paper is organized as follows. Section 2 reviews the literature on the impacts of recreational marijuana legalization, including general societal effects, consequences for secondary education, and emerging evidence on postsecondary outcomes. Section 3 provides background on marijuana policy in the United States, Washington State's legalization framework, and the state's randomized dispensary license lottery. Section 4 describes the institutional data, treatment assignment strategy, and key variables. Section 5 outlines the empirical methodology, with an emphasis on the staggered difference-in-differences design. Section 6 presents the main results, and Section 7 conducts robustness checks across alternative treatment thresholds. Section 8 discusses the findings in light of behavioral mechanisms and related literature. Section 9 concludes with implications for policy and future research.

2 Literature Review

2.1 Recreational Marijuana Legalization: General Impacts

Since the early 2010s, a growing body of empirical research has assessed the broad consequences of recreational marijuana legalization (RML) across health, social, and economic

domains. Early-legalizing states like Washington and Colorado have served as policy laboratories for evaluation. Contrary to worst-case fears, much of the evidence to date suggests that RML has not produced widespread negative societal outcomes.

Several studies document a rise in adult cannabis use post-legalization without corresponding increases among adolescents. Using repeated cross-sectional data from the Washington Young Adult Health Survey, Kilmer et al. (2022) find that cannabis consumption increased among adults aged 21 and older but remained stable among adolescents. Similarly, Bailey et al. (2023) analyze three longitudinal cohort studies across different states and apply multilevel models of repeated measures, concluding that cannabis use among adolescents did not significantly rise after legalization.

Beyond usage trends, RML has produced broader social and economic effects. Everson et al. (2019), using logistic regression models on Washington BRFSS data linked to local cannabis retail openings, find that proximity to cannabis stores increased adult cannabis use but was not associated with higher rates of problematic use or health harms.

Meta-analytical evidence further supports these findings. Chan et al. (2024) conduct a systematic review and meta-analysis, showing that while adolescent cannabis use can negatively affect academic achievement, the magnitudes are heterogeneous and context-dependent.

Overall, these studies suggest that RML has increased adult cannabis access and normalized usage patterns without precipitating a societal crisis.

2.2 Impacts on Secondary Education

More recent research has shifted toward understanding RML’s impacts on adolescent educational outcomes. A prominent contribution is Jarrold-Grapes (2023a), who exploits exogenous variation from Washington State’s dispensary license lottery. Using an instrument variable framework, she compares high schools within 10 minutes’ driving distance of a dispensary to more distant schools. She instruments for a school’s proximity to an open dispensary with its proximity to a lottery winner. Her findings show substantial increases in dropout rates and chronic absenteeism among treated students, especially among girls, though standardized test scores were not significantly affected.

These results are consistent with neuroscience studies indicating adolescent females may be more sensitive to THC’s effects on emotion regulation and executive function (Belfiore et al., 2024; Kumar et al., 2024).

In parallel, Jarrold-Grapes (2023b) examines Oregon’s legalization using a different approach. Exploiting municipal variation in cannabis dispensary opt-in status, she applies an instrumental variables strategy similar to her paper on Washington schools. Her findings also suggest modest negative effects on school outcomes, although the magnitudes are smaller compared to Washington.

However, when viewed at the macro level, high school educational attainment has continued to improve post-legalization. Mosher and Akins (2020) use longitudinal administrative data to show that Washington State’s high school graduation rates rose even after RML, likely reflecting broader systemic reforms. Similarly, Dills et al. (2021) apply synthetic con-

trol methods across multiple states and conclude that legalization had no detectable adverse effect on overall high school graduation rates.

Thus, while localized harms exist, aggregate educational outcomes appear resilient to legalization at the state level.

2.3 Impacts on Postsecondary Education

Research on the postsecondary sector is more limited but growing. International evidence from Marie and Zölitz (2017) leverages a natural experiment in Maastricht, Netherlands, where cannabis access was suddenly restricted based on nationality. Using a difference-in-differences design with administrative university records, Marie and Zölitz find that restricting cannabis access significantly improved students’ academic performance, especially in challenging courses like mathematics and engineering.

In the United States, scholars have investigated RML’s effects on college applications and enrollment. Blake et al. (2024) use a two-way fixed effects difference-in-differences model to show that legalization increased college applications to public universities in legalizing states by about 5%, without reducing applicant quality. Similarly, El Fatmaoui (2024) applies an event-study framework and finds that RML increased college enrollment by 5–9%, primarily among out-of-state students. Notably, El Fatmaoui finds no evidence that legalization adversely impacted graduation rates or academic performance.

However, most of these studies focus on state-level effects and overlook within-state variation in cannabis access. Everson et al. (2019) emphasize that adult behavior is highly sensitive to local cannabis availability, suggesting that analyses aggregating by state may mask important local effects.

Taken together, the literature indicates that while RML has not undermined postsecondary academic success broadly, more research is needed to understand the local educational consequences of cannabis market expansion.

3 Background

3.1 U.S. Marijuana Laws

Cannabis policy in the United States has undergone a dramatic transformation over the past two decades. While marijuana remains illegal under federal law—classified as a Schedule I substance under the Controlled Substances Act—many states have adopted more permissive regimes. As of 2024, 24 states and the District of Columbia have legalized recreational marijuana use for adults aged 21 and over, and more than 30 states allow medical marijuana in some form (National Conference of State Legislatures, 2024). This decentralized approach has resulted in a patchwork of laws across the country, enabling researchers to examine the localized effects of legalization.

State-level reforms typically begin with the decriminalization of possession, followed by the introduction of medical marijuana programs, and eventually the legalization of adult-use cannabis. Although federal prohibition creates legal uncertainty, the Department of

Justice under the Obama administration issued guidance (e.g., the 2013 Cole Memorandum) indicating that federal enforcement would not interfere with state-legal cannabis activity provided certain priorities (e.g., preventing sales to minors or diversion to other states) were met.

3.2 Washington’s Recreational Marijuana Law

Washington State was one of the first jurisdictions to legalize recreational marijuana. In November 2012, voters approved Initiative 502 (I-502), legalizing the possession and use of cannabis for adults aged 21 and over. The law went into effect on December 6, 2012, and legalized not only possession (up to one ounce) but also authorized the state to regulate the production, processing, and sale of marijuana through licensed businesses.

I-502 directed the Washington State Liquor Control Board (now the Washington State Liquor and Cannabis Board, WSLCB) to design and implement a regulatory framework for cannabis cultivation and retail distribution. After a detailed rulemaking process, the state began accepting applications in 2013, and the first recreational marijuana stores opened in July 2014.

Washington’s system included strict zoning, product testing, packaging, and advertising requirements to limit youth access and ensure product safety. Importantly, retail sales were subject to substantial excise taxes, with revenues earmarked for public health, education, substance abuse prevention, and local governments. These features make Washington’s system one of the most regulated and heavily studied in the country (Hansen et al., 2020).

3.3 Washington’s Marijuana Retail License Lottery

To allocate retail licenses under I-502, the WSLCB implemented a lottery-based system. After receiving more than a thousand applications for just 334 retail licenses across the state, the WSLCB used a third-party vendor to conduct a random drawing in May 2014. Each applicant had to submit a specific address for their proposed store location in advance, and successful applicants were selected within jurisdiction-specific quotas (e.g., counties and cities had a limited number of slots). Applicants also had to meet background check and business requirement criteria to qualify.

This randomized mechanism created quasi-experimental variation in the placement of retail cannabis stores, which has been leveraged by researchers to study causal effects on local outcomes such as crime (Dong and Tyndall, 2024), property values, and now, educational outcomes. Since the lottery created variation in dispensary access that was not directly tied to local demand or neighborhood characteristics, it provides a compelling identification strategy for isolating the localized impacts of legal cannabis availability.

4 Data

This study combines data from multiple sources to examine how the proximity of recreational marijuana dispensaries affects postsecondary educational outcomes in Washington State.

The following section describes the data sources, construction of key variables, and the process of assigning treatment and control groups.

4.1 Lottery Results and Dispensary Openings

Information on marijuana dispensary licensing and openings was obtained from the Washington State Liquor and Cannabis Board (WSLCB). The state conducted lotteries in 75 jurisdictions to allocate dispensary licenses, while 47 jurisdictions did not require a lottery. In total, there were 1,176 applicants to the dispensary licensing process, among whom 253 won and 875 lost. However, not all winners proceeded to open dispensaries. Of the 253 winners, 177 dispensaries eventually opened during the study period between 2014 and 2017, while the rest did not.¹ Figure 1 shows the distribution of the winners and losers of the lottery. It is important to note that even though dispensary applicants were concentrated in certain areas, their location appears to be mostly randomized within that area because of the lottery system.

In addition to licensing data, monthly sales and tax records from WSLCB were used to track the opening and closing dates of dispensaries over time. These dynamic indicators allow the identification of which dispensaries were operational in a given year.

Figure 2 illustrates the operational status of each recreational dispensary in Washington by month for the first two years, with each dispensary represented by a line (active period) from its opening to closing date. The figure shows a steady entry of dispensaries into the market after legalization: new dispensaries opened nearly every month from mid-2014 onward, and once opened, most remained in operation through the end of my study period (closures were rare). This expansion of dispensaries means that the number of campuses exposed to a nearby dispensary—and the duration of exposure—increased over time. In other words, treatment intensity is not static; early in the post-legalization period, relatively fewer colleges had dispensaries in their vicinity, whereas later on many more did. This observation suggests that a binary treatment indicator may oversimplify the exposure. Future analyses might incorporate measures such as the count of dispensaries within a given radius as alternative treatment. For now, I note that the gradual ramp-up of dispensaries provides useful variation that my DiD model can exploit in identifying effects.

4.2 College Data

Data on postsecondary institutions were obtained from the Integrated Postsecondary Education Data System (IPEDS), maintained by the U.S. Department of Education. The analytic sample consists of 127 unique postsecondary institutions located in Washington State, covering academic years from 2008 to 2017. This nine-year window allows for the examination of trends both before and after the legalization and retail rollout of recreational marijuana. Specifically, the dataset includes six years of pre-treatment data (2008–2013) and three years of post-treatment data (2014–2016), corresponding to the period following the implementation of Washington’s Initiative 502 and the opening of licensed dispensaries.

¹There is some discrepancy in the numbers as the winners and losers don’t add up to the total application numbers. This most likely is due to reporting or minor documenting inconsistencies by the WSLCB.

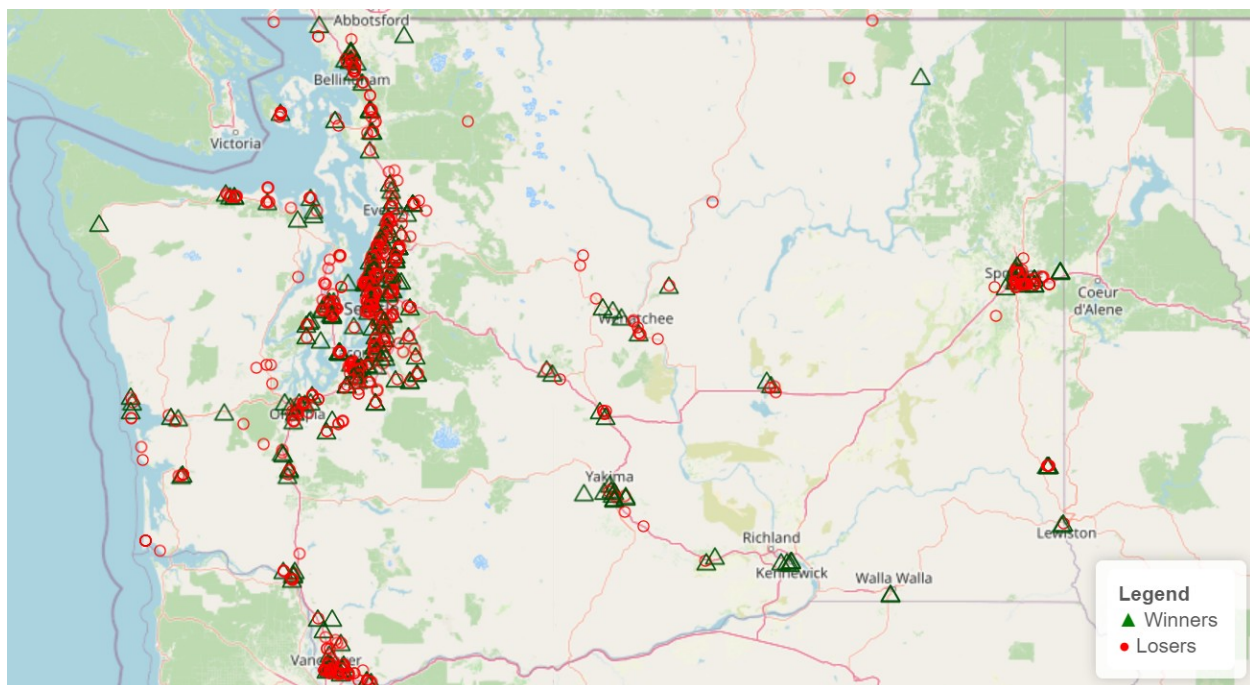


Figure 1: Lottery Winners and Losers

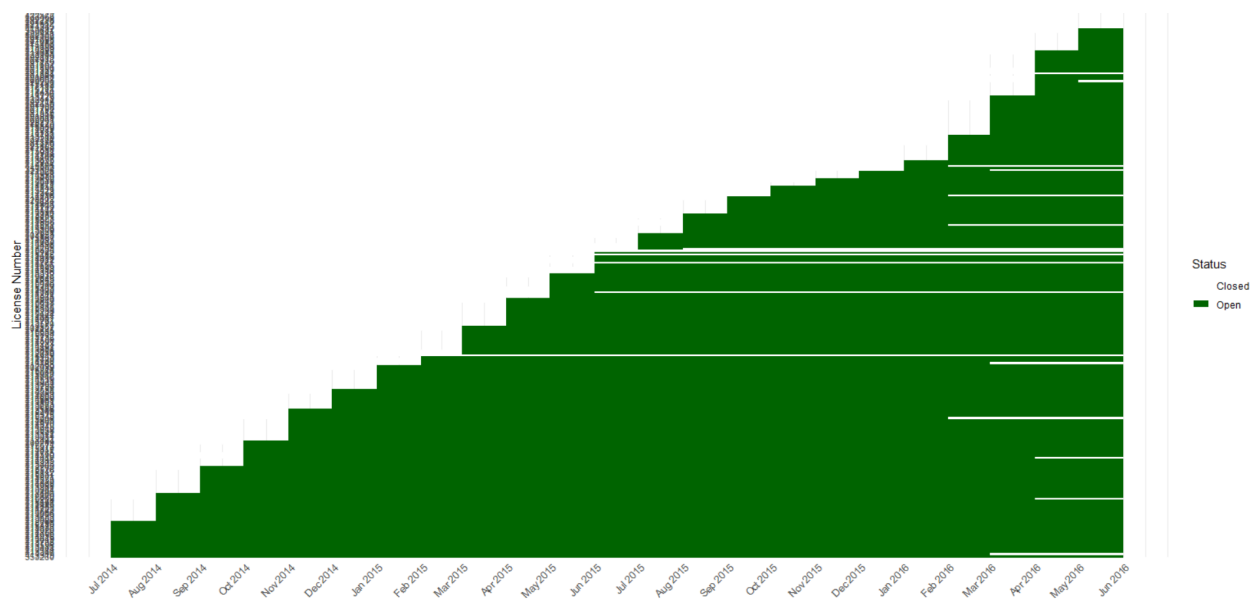


Figure 2: Dispensary Operational Status Over Time

A key feature of the IPEDS data structure is that it includes disaggregated entries for institutions with multiple physical locations. That is, branch campuses, satellite colleges, and extension centers are recorded as separate data points when they operate independently and report outcomes at the campus level. This distinction is important for the present study, as it enables the use of localized treatment assignment based on the geographic proximity of each campus to marijuana dispensaries. As a result, the analysis reflects true spatial variation in institutional exposure rather than aggregating across potentially heterogeneous locations within a single system.

The primary outcome variables are undergraduate enrollment and student retention rates, both reported annually at the institution level. Undergraduate enrollment captures the total number of degree- or certificate-seeking undergraduates enrolled during the fall census date, while retention measures the percentage of first-time, full-time students returning to the same institution in the following academic year.

In addition to outcome measures, several institutional characteristics are included as covariates to account for potential confounders. These include the level of the institution (2-year vs. 4-year), total undergraduate enrollment (used to categorize institutional size), sector (public, private not-for-profit, or private for-profit), and locale classifications (urban, suburban, town, or rural), which are derived from U.S. Census definitions and included in IPEDS. These variables are used to control for baseline differences across institutions that might influence both their exposure to dispensaries and educational outcomes.

4.3 Proximity Measures and Treatment Assignment

Constructing a credible measure of institutional exposure to recreational marijuana access involved multiple steps, ranging from data gathering to spatial computation. The ultimate goal was to determine which universities were within a specific driving-time threshold of an operational marijuana dispensary in a given year, and to assign treatment status accordingly.

The process began by collecting monthly tax records from the Washington State Liquor and Cannabis Board (WSLCB), which provided information on sales activity for each dispensary. This data was used to infer which dispensaries were operational in each month by identifying when a dispensary began reporting taxable sales and for how long. Based on this information, a list of dispensaries and their respective opening dates was constructed.

Next, the physical addresses of these dispensaries were submitted to the Google Geolocation API to retrieve geographic coordinates (latitude and longitude). These coordinates were then plotted on a map to visualize the spatial distribution of dispensaries across the state. Similarly, the locations of all universities in Washington were geocoded and plotted on the same map.

To reduce the computational cost associated with querying the Google Distance Matrix API, an initial filtering step was applied using the Haversine formula in R. This method calculated the linear distance between each university-dispensary pair. Pairs that exceeded a threshold of 48 kilometers—roughly equivalent to a 30-minute drive—were excluded from further analysis, as they would be unlikely to fall within the eventual treatment window.

The remaining filtered pairs were then passed through the Google Distance Matrix API,

which returned the estimated driving time between each university and each dispensary. These values served as the basis for treatment assignment. A university was designated as treated in a given year if it was located within a 10-minute driving radius of a dispensary that had opened by that year. Universities beyond this threshold were classified as control institutions. Importantly, the year in which a university first came within the 10-minute threshold was recorded as the timing of treatment, allowing for dynamic modeling of exposure effects over time.

5 Empirical Methodology

This study exploits the staggered timing of marijuana dispensary openings in Washington State to identify the causal effect of local cannabis access on college enrollment and retention rates. I use a difference-in-differences (DiD) framework adapted for multiple time periods and staggered treatment adoption, following the estimator developed by Callaway and Sant’Anna (2021). This framework explicitly accommodates treatment effect heterogeneity across groups and over time and avoids the biases associated with conventional two-way fixed effects (TWFE) estimators (Goodman-Bacon, 2021; Sun and Abraham, 2021).

Let Y_{it} denote the outcome of interest (e.g., first-time enrollment or one-year retention) for unit i at time t . Each unit is either untreated or treated depending on whether a marijuana dispensary has opened in proximity. Let G_i denote the period in which unit i is first exposed to treatment (i.e., when a dispensary opens nearby). Units that are never treated are assigned $G_i = \infty$. The treatment status of unit i at time t is given by the indicator $D_{it} = \mathbb{1}\{t \geq G_i\}$.

Under the potential outcomes framework, define $Y_{it}(1)$ as the potential outcome for unit i at time t if treated, and $Y_{it}(0)$ as the potential outcome in the absence of treatment. The observed outcome is therefore:

$$Y_{it} = D_{it} \cdot Y_{it}(1) + (1 - D_{it}) \cdot Y_{it}(0). \quad (1)$$

The causal effect of interest is the average treatment effect on the treated (ATT) for units first treated in period g , observed in period t . This is referred to as the group-time ATT:

$$ATT(g, t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid G_i = g]. \quad (2)$$

This quantity captures the average effect of treatment at time t for the cohort of units first treated in period g . Notably, $ATT(g, t)$ is only defined for $t \geq g$.

The key identifying assumption is the conditional parallel trends assumption, which states that in the absence of treatment, the average potential outcome under control for treated and not-yet-treated units would evolve similarly:

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid G_i = g] = \mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0) \mid G_i > t], \quad (3)$$

for all $t \geq g$. This assumption is weaker and more flexible than the unconditional parallel trends assumption imposed in standard DiD models.

To estimate the group-time ATT, I use the 'doubly robust estimator' of Callaway and Sant'Anna (2021), which combines outcome regression and inverse probability weighting. This approach improves efficiency and robustness to model misspecification. Group-time ATT estimates are then aggregated across cohorts to form overall average treatment effects (ATTs) or dynamic treatment effects by event time $\tau = t - g$.

I estimate the following event-study regression:

$$Y_{it} = \sum_{\tau \neq -1} \theta_{\tau} \cdot \mathbb{1}\{t - G_i = \tau\} + \gamma_i + \lambda_t + \varepsilon_{it}, \quad (4)$$

where θ_{τ} represents the average treatment effect at event time τ , γ_i are unit fixed effects, and λ_t are time fixed effects. The period just prior to treatment ($\tau = -1$) is omitted and serves as the baseline. Estimates of θ_{τ} provide a dynamic view of the treatment effect relative to the pre-treatment period.

I assume irreversibility of treatment—once a unit is treated, it remains treated thereafter. I also assume no anticipation, meaning that treatment has no effect on outcomes prior to actual treatment implementation. Robustness to this assumption is assessed by testing for the presence of statistically significant leads in the event-study specification.

All estimation is implemented using the `did` package in R, which operationalizes the Callaway and Sant'Anna estimator. Inference is based on the multiplier bootstrap method, which provides valid simultaneous confidence intervals for treatment effects across multiple time periods.

5.1 Baseline Characteristics

Table 1 summarizes the baseline (pre-legalization (2013)) characteristics of the treatment and control institutions. I compare key covariates including institution type (four-year vs. two-year), size, locale (urban, suburban, rural, town), and sector (public vs. private, non-profit vs. for-profit). Overall, the two groups are similar along most dimensions. Differences in sector, control, institution level, and size are small and not statistically significant. One notable difference is in locale: a higher proportion of treated institutions are located in suburban areas compared to control institutions, and this disparity is significant at the 5% level. This imbalance suggests that locale (especially being in a suburban setting) may be correlated with treatment status. In the subsequent analysis, I account for such differences by including locale fixed effects or other controls to ensure an apples-to-apples comparison between treated and control institutions. Aside from this, the generally balanced nature of other covariates supports the validity of my treatment-control comparison.

6 Results

6.1 Effect on Student Retention

The primary outcome of interest is the student retention rate, defined as the proportion of students who return to the same institution in the subsequent academic year.

	Proportion of Institutions			
	Treatment	Control	Difference	P-value
Panel A: Institution Level				
2–4 year college	0.3438	0.3548	-0.0111	0.9122
4+ year college	0.4271	0.4194	0.0077	0.9407
≤2 year college	0.2292	0.2258	0.0034	0.9696
Panel B: Institution Size				
Large (>20k students)	0.0104	0.0323	-0.0218	0.5234
Medium (5k–20k)	0.1250	0.2903	-0.1653	0.0722
Small (<5k)	0.8125	0.6774	0.1351	0.1589
Panel C: Locale				
City	0.6771	0.6771	-0.0004	0.9973
Rural	0.0625	0.0968	-0.0343	0.2565
Suburb	0.2604	0.0968	0.1636	0.0226
Town	0.0313	0.1290	-0.0978	0.1340
Panel D: Sector/Control				
Private for-profit	0.4167	0.4516	-0.0349	0.7383
Private not-for-profit	0.2604	0.1290	0.1313	0.0886
Public	0.3229	0.4194	-0.0964	0.3495

Table 1: Baseline Characteristics

I begin by examining the raw trends in student retention rates for treated versus control institutions. Figure 3 plots the average retention rate over time for the two groups. In the years prior to legalization (pre-2014), treated and control colleges exhibited roughly parallel retention trajectories, with year-to-year fluctuations but no clear divergence between the groups. After 2014, however, a slight gap emerges: control institutions (those without a nearby dispensary) show a modest increase in retention rates, whereas the treated institutions experience a flatter trend in the immediate post-legalization years, followed by a gradual uptick later on. Overall, retention rates in both groups hover in the 70–80% range throughout the sample period. These descriptive patterns hint at a possible negative impact of dispensary proximity on retention (as the treatment group lags behind slightly in the post-legalization period). Note that this graph is only illustrative, as the control and treatment group is actually dynamic since dispensaries open gradually each year.

Next, using a staggered difference-in-differences design, I find strong evidence that proximity to recreational marijuana dispensaries negatively affects retention. Specifically, institutions located within a 10-minute driving radius of a licensed dispensary experienced an average decline in retention rates of 7.9 percentage points, as shown in Table 2. This effect is statistically significant and robust to alternative specifications and covariate controls. The group-time ATT estimates reveal that the effect becomes more pronounced after one to two academic years of dispensary exposure.

Figure 5 presents the dynamic treatment effects by event time. The absence of significant pre-treatment effects provides empirical support for the parallel trends assumption. The treatment effect intensifies in post-treatment periods, suggesting that the impact accumulates

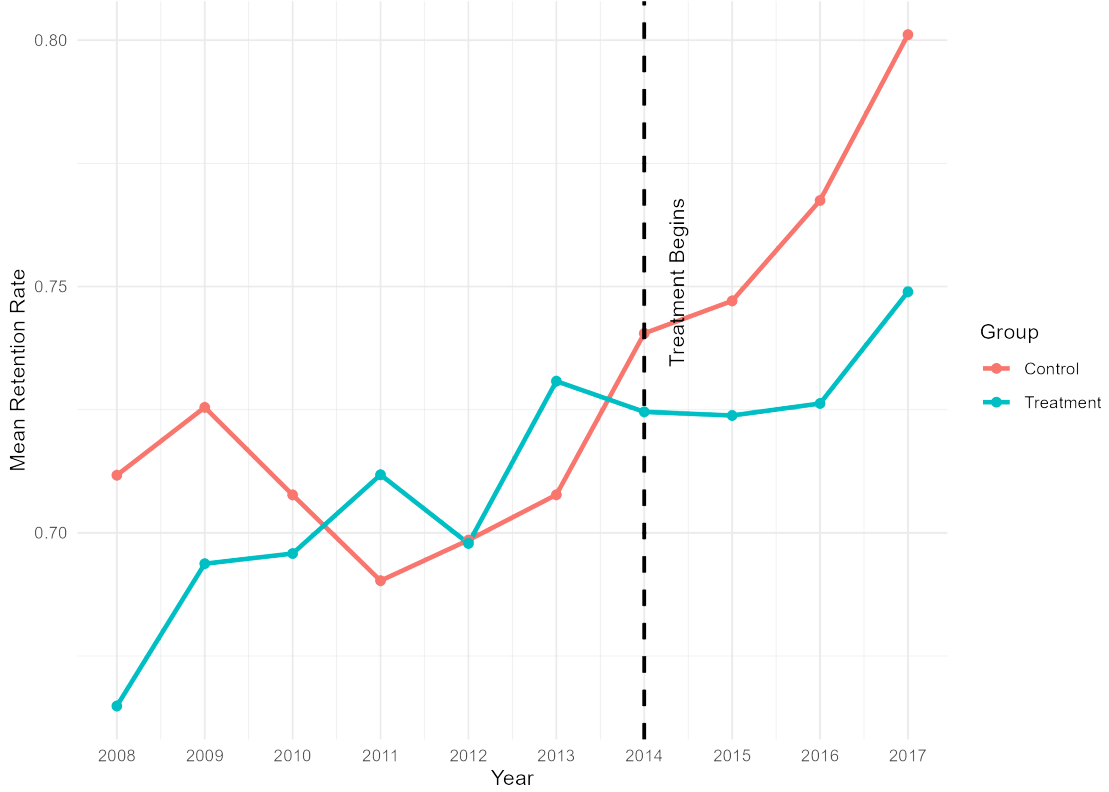


Figure 3: Trends in Student Retention Rates by Treatment Group

over time rather than arising from short-term shocks.

6.2 Effect on Enrollment

In contrast to the decline in retention, I find no evidence that proximity to a dispensary significantly alters college enrollment rates as reported in Table 4. Enrollment trends remained relatively stable across both treated and control institutions during the pre- and post-treatment periods.

Figure 5 displays the estimated dynamic effects on enrollment, showing consistently small and statistically insignificant coefficients across the full event-time horizon.

7 Robustness Check

To assess the robustness of the estimated treatment effects, I vary the definition of treatment by adjusting the maximum driving-time threshold used to define dispensary proximity. The main specification uses a 10-minute threshold; however, I re-estimate the average treatment effects on student retention using thresholds ranging from 5 to 15 minutes.

Table 7 summarizes the ATT estimates for each threshold. The results show a consistent pattern: treatment effects are small and statistically insignificant at very short distances

Overall Summary of ATT Based on Event-Study/Dynamic Aggregation			
ATT	Std. Error	[95% Conf. Int.]	
-0.079	0.0298	[-0.1373, -0.0207]*	
Signif. codes: ‘*’ confidence band does not cover 0			
Dynamic Effects			
Event time	Estimate	Std. Error	95% Simult. Conf. Band
-6	-0.0293	0.0361	[-0.1204, 0.0619]
-5	-0.0106	0.0259	[-0.0760, 0.0547]
-4	0.0802	0.0403	[-0.0215, 0.1819]
-3	-0.0203	0.0325	[-0.1022, 0.0616]
-2	-0.0287	0.0356	[-0.1184, 0.0609]
-1	-0.0029	0.0252	[-0.0665, 0.0606]
0	-0.0243	0.0186	[-0.0712, 0.0227]
1	-0.0878	0.0370	[-0.1811, 0.0056]
2	-0.1081	0.0503	[-0.2349, 0.0188]
3	-0.0959	0.0446	[-0.2082, 0.0165]
Estimation Details			
Control Group:	Not Yet Treated		
Anticipation Periods:	0		
Estimation Method:	Doubly Robust		

Table 2: Difference-in-Differences Analysis Results for Retention Rates

(5–9 minutes), but become larger and statistically significant at thresholds of 10 minutes and beyond.

Specifically, the ATT estimate at the 10-minute threshold is -0.079 with a 95% confidence interval of $[-0.1352, -0.0228]$, and remains significant through 15 minutes. The magnitude of the effect remains relatively stable between 10 and 15 minutes, suggesting that the treatment effect is not sensitive to modest changes in threshold definition and likely captures a meaningful exposure effect due to dispensary proximity.

Since the ATT estimates are weighted by institution size, one concern is that the results may be disproportionately driven by a small number of large universities. To address this, I conduct a robustness check in which the largest institutions—based on total undergraduate enrollment—are removed from the sample. The re-estimated effects are substantively unchanged, suggesting that the results are not an artifact of outlier weighting and reflect broader institutional trends.

Importantly, the effect on enrollment rates remains statistically insignificant across all proximity thresholds, as shown in Table 5. This strengthens the conclusion that dispensary access does not influence students’ initial enrollment decisions but may instead impact their academic persistence once they are enrolled in college.

These results reinforce the main findings and suggest that the observed decline in student retention is robust to reasonable variations in the treatment assignment rule. Moreover, the pattern aligns with the idea that dispensary presence must be within a practical distance

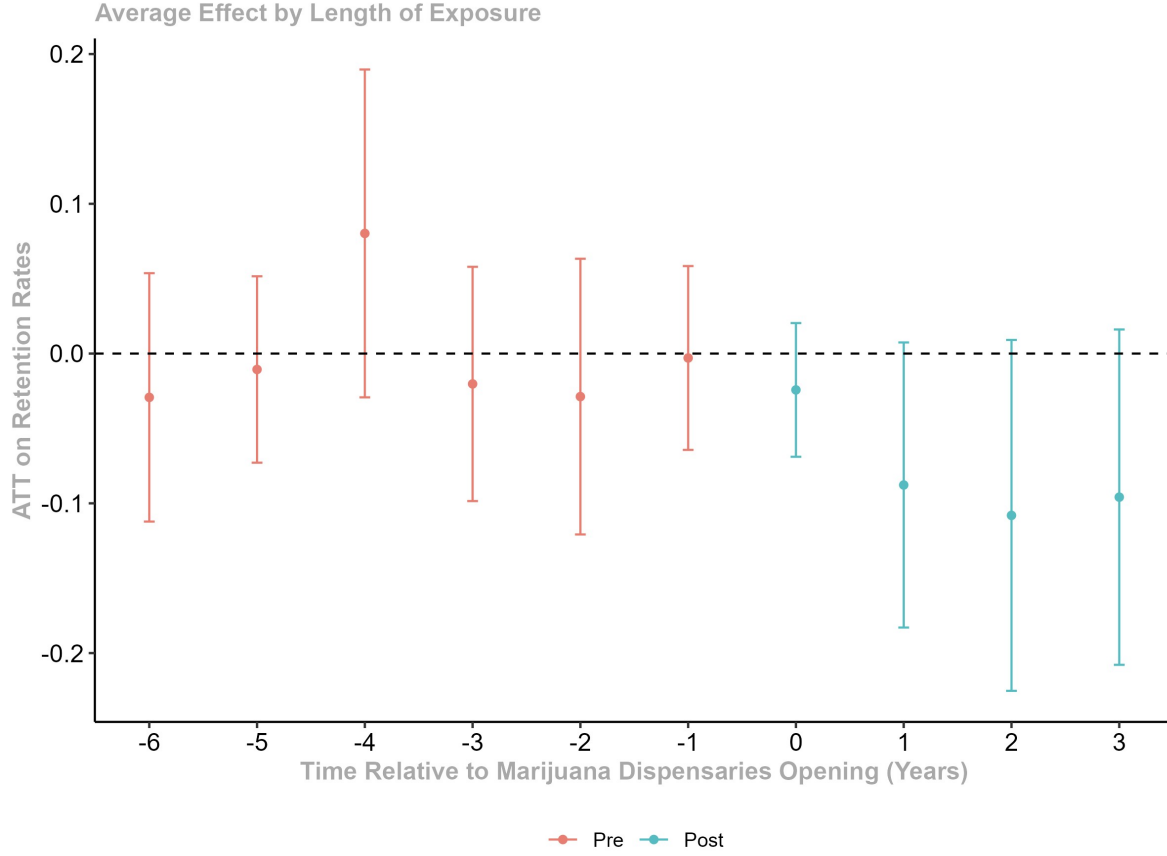


Figure 4: Event Study Estimates for Student Retention Rates

to influence behavior, consistent with prior research on localized retail cannabis impacts (Everson et al., 2019).

Distance Threshold (min)	ATT	Std. Error	95% Confidence Interval
5	-0.0125	0.0254	[-0.0622, 0.0372]
6	-0.0381	0.0241	[-0.0852, 0.0091]
7	-0.0478	0.0266	[-0.0999, 0.0043]
8	-0.0408	0.0239	[-0.0876, 0.0060]
9	-0.0257	0.0297	[-0.0839, 0.0326]
10	-0.0790	0.0287	[-0.1352, -0.0228]
11	-0.0785	0.0299	[-0.1371, -0.0200]
12	-0.0666	0.0272	[-0.1200, -0.0132]
13	-0.0666	0.0283	[-0.1221, -0.0110]
14	-0.0779	0.0324	[-0.1414, -0.0143]
15	-0.0779	0.0354	[-0.1472, -0.0085]

Table 3: Robustness Check: ATT Estimates for Student Retention by Proximity Threshold

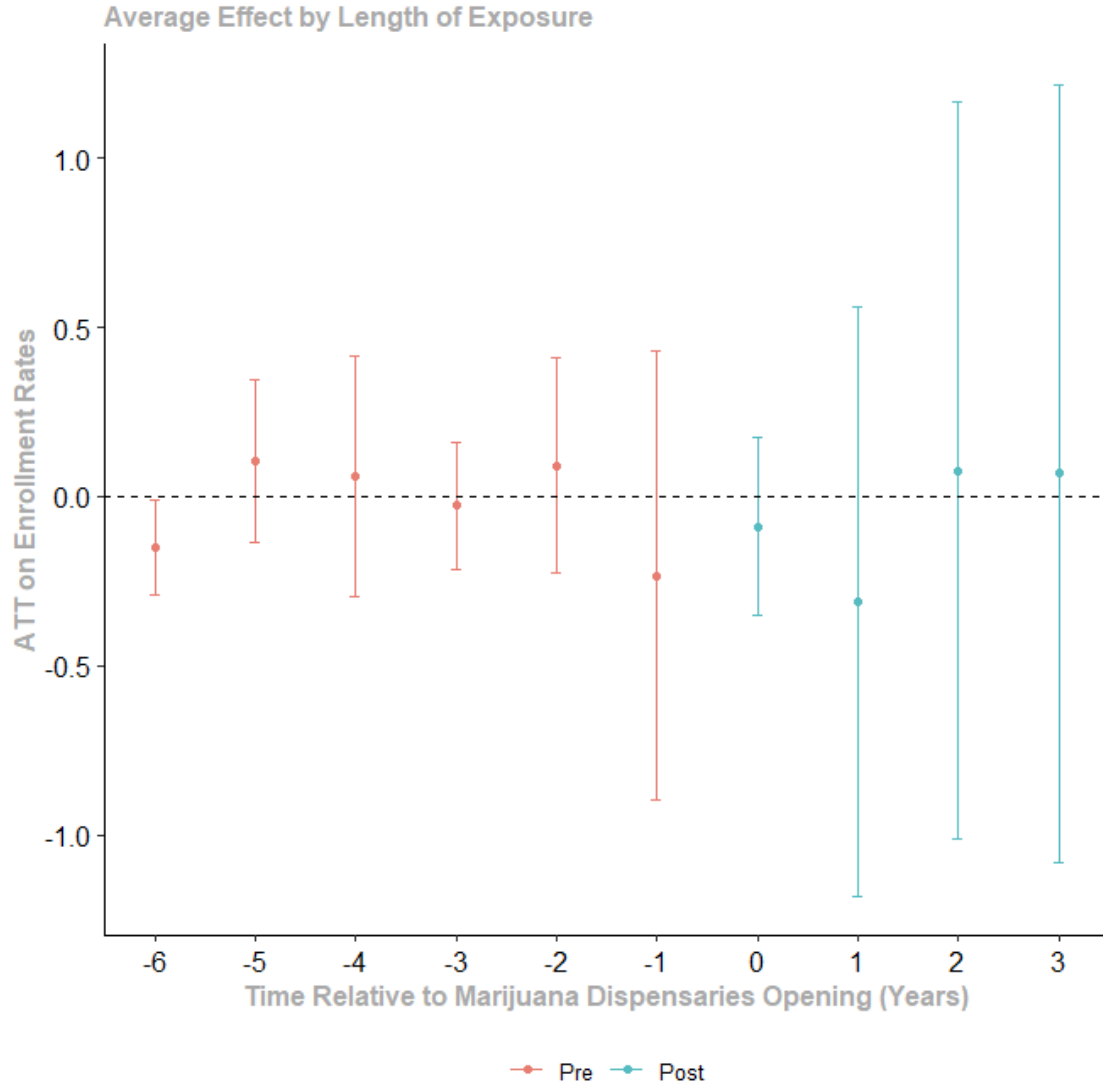


Figure 5: Event Study Estimates for Enrollment Rates

8 Discussion

The empirical findings from this study reveal a striking asymmetry in the educational effects of recreational marijuana legalization: while initial college enrollment remains unaffected, student retention experiences a significant and persistent decline in response to dispensary proximity. This suggests that the academic risks associated with cannabis legalization may not materialize during the application or matriculation phase, but rather emerge during the course of a student's academic progression.

One potential mechanism is increased cannabis consumption due to reduced acquisition costs. Proximity to a retail dispensary may lower both logistical and psychological barriers to access, potentially increasing usage among enrolled students. Although this study does not directly observe individual consumption patterns, prior research provides supporting

evidence for this channel. For example, Everson et al. (2019) find that adult cannabis use in Washington State rose substantially following the opening of nearby dispensaries, even after accounting for local demographic variation. These findings imply that physical access is a strong determinant of use, particularly in walkable or student-dense settings, and it is plausible that similar patterns hold among college students.

This behavioral channel is further supported by clinical literature linking cannabis use to diminished academic performance. Arria et al. (2015), for instance, report that frequent users exhibit lower GPAs, increased absenteeism, and weaker academic persistence. Even if overall adolescent usage rates remain stable post-legalization, the increased accessibility may lead to higher intensity or frequency of use among students at risk of attrition.

Beyond direct consumption, dispensary proximity may shape educational outcomes through broader environmental mechanisms. The visibility and normalization of marijuana in a campus’s immediate vicinity could influence peer norms, time allocation, or perceived institutional expectations. These environmental factors may not affect college enrollment decisions—often driven by broader reputational and geographic considerations—but could erode student motivation or academic discipline after matriculation.

Importantly, robustness checks varying the definition of treatment further reinforce these findings. When redefining treatment by adjusting the maximum driving-time threshold from 5 to 15 minutes, the adverse effect on retention is consistently significant for thresholds of 10 minutes and beyond, and remains stable in magnitude. This pattern indicates that the observed effect is not an artifact of arbitrary treatment definition, but reflects meaningful behavioral exposure due to dispensary proximity. Conversely, the null results on enrollment persist across all threshold values, strengthening the conclusion that local cannabis access does not influence initial college choice.

These patterns are consistent with findings from secondary education. Jarrold-Grapes (2023a) document that high school students in Washington exposed to nearby dispensaries show increased dropout and absenteeism, particularly among upper-grade cohorts. The presence of similar dynamics in the postsecondary setting underscores the importance of considering localized spillovers even when broader educational indicators appear unaffected.

An important limitation of the analysis is the lack of direct data on the number of residential (on-campus) students at each institution. It is plausible that the effects of dispensary proximity are more pronounced at institutions where a larger share of the student body lives on or near campus, due to higher likelihood of localized exposure. However, consistent data on residential enrollment at the campus level are not available through IPEDS or other administrative sources for the full sample.

Despite this limitation, several features of the research design mitigate concern that the results are entirely driven by a small number of highly residential campuses. First, the main analysis is weighted by total undergraduate enrollment, which already places greater emphasis on institutions with large student populations. Second, the robustness check that excludes the largest institutions (many of which are also more residential) shows that the treatment effect on retention remains stable. Third, the randomized nature of dispensary openings and the geographically defined treatment help reduce bias from unobserved campus characteristics.

Nevertheless, future work could explore how cannabis access interacts with campus residential intensity, perhaps by combining institutional data with student housing statistics or geolocation-based exposure proxies.

Taken together, these findings indicate that recreational dispensary access may have non-trivial consequences for academic persistence, particularly in localized educational environments. The results also highlight the need for policy approaches that consider geographic nuance, rather than relying solely on state-level aggregates or average effects.

9 Conclusion

This paper examines how the proximity of recreational marijuana dispensaries affects post-secondary educational outcomes in Washington State. Leveraging the quasi-random assignment of retail licenses via a state-run lottery and a staggered difference-in-differences design, I find robust evidence that localized dispensary access leads to a significant decline in student retention, while having no measurable effect on college enrollment.

These findings are reinforced by a series of robustness checks that vary the treatment definition across a range of proximity thresholds. The estimated retention effects remain stable and statistically significant for distances of 10 to 15 minutes, indicating that the results are not sensitive to reasonable changes in treatment assignment rules. The enrollment effects, by contrast, remain consistently null regardless of specification.

Together, these results have several implications. Academically, they underscore the importance of analyzing localized policy spillovers and caution against assuming uniform impacts of cannabis legalization across student populations or institutions. From a policy standpoint, the findings suggest that zoning laws and campus-adjacent regulations may be warranted to mitigate adverse educational consequences. Moreover, institutions in high-exposure areas may consider strengthening student support services or wellness initiatives targeted at substance use.

Future research should investigate heterogeneity by student characteristics and institutional composition, and should explore long-term outcomes such as graduation rates and academic performance. Further work linking institutional data with behavioral and health records could help isolate the underlying mechanisms driving attrition and inform targeted interventions in the era of widespread cannabis legalization.

References

- Anderson, D. M. and Rees, D. I. (2014). The legalization of recreational marijuana: how likely is the worst-case scenario? *Journal of Policy Analysis and Management*, 33(1):221.
- Arria, A. M., Caldeira, K. M., Vincent, K. B., Winick, E. R., Baron, R. A., and O’Grady, K. E. (2015). Marijuana use patterns and sleep among college students: A longitudinal study. *Journal of Studies on Alcohol and Drugs*, 76(1):73–81.
- Bailey, J. A., Tiberio, S. S., and Kerr, D. C. (2023). Effects of cannabis legalization on adolescent cannabis use across 3 studies. *American Journal of Preventive Medicine*, 64(2):173–181.
- Becker, G. S. and Murphy, K. M. (2013). Have we lost the war on drugs? *Wall Street Journal*, 4:A1.
- Belfiore, C., Galofaro, V., Cotroneo, D., Lopis, A., Tringali, I., Benfante, R., Allegra, M., and Tiziana, C. (2024). A multi-level analysis of biological, social, and psychological determinants of substance use disorder and co-occurring mental health outcomes. *Psychoactives*, 3(2):13.
- Blake, C. D., Thomas, D. K., and Hess, J. (2024). Higher education: The impact of recreational marijuana on college applications. *Contemporary Economic Policy*, 42(2):259–277.
- Callaway, B. and Sant’Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.
- Chan, O., Daudi, A., Ji, D., Wang, M., Steen, J., Parnian, P., Li, C., Xiong, A., Zhang, W., Lopes, L., MacKillop, J., Busse, J., and Wang, L. (2024). Cannabis use during adolescence and young adulthood and academic achievement: A systematic review and meta-analysis. *JAMA Pediatrics*, 178(12):1280–1289.
- Dills, A. K., Goffard, S., and Miron, J. (2021). The effect of state marijuana legalizations: 2021 update. Technical Report Report 900, Cato Institute Policy Analysis.
- Dong, X. (2022). Recreational marijuana sales legalization and monday work injury claims. *The BE Journal of Economic Analysis & Policy*, 22(1):99–121.
- Dong, X. and Tyndall, J. (2024). The impact of recreational marijuana dispensaries on crime: Evidence from a lottery experiment. *The Annals of Regional Science*, 72:1383–1414.
- El Fatmaoui, A. (2024). From high school to higher education: Is recreational marijuana a consumption amenity for US college students? *Economic Inquiry*, 62(3):1024–1045.
- Everson, E. M., Dilley, J. A., and Maher, J. E. (2019). Post-legalization opening of retail cannabis stores and adult cannabis use in washington state, 2009–2016. *American Journal of Public Health*, 109(9):1292–1299.
- Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

- Hansen, B., Miller, K., and Weber, C. (2017). *The grass is greener on the other side: How extensive is the interstate trafficking of recreational marijuana?* National Bureau of Economic Research.
- Hansen, B., Miller, K., and Weber, C. (2020). The taxation of recreational marijuana: Evidence from washington state. *American Economic Journal: Economic Policy*, 13(2):157–190.
- Jarrold-Grapes, R. (2023a). Does recreational marijuana legalization harm students? evidence from washington state’s dispensary lottery. Working Paper, Federal Trade Commission.
- Jarrold-Grapes, R. (2023b). Marijuana legalization and educational outcomes: Evidence from oregon. Working Paper.
- Kilmer, J. R., Rhew, I. C., Guttmanova, K., and Fleming, C. B. (2022). Cannabis use among young adults in washington state after legalization of nonmedical cannabis. *American Journal of Public Health*, 112(4):587–596.
- Kumar, J., Mohamed, W., Borroto-Escuela, D., Fuxe, K., and Rizvi, S. (2024). Substance use disorder: Above and beyond addiction. *Frontiers in Pharmacology*, 15:1426351.
- Marie, O. and Zölitz, U. (2017). High achievers? cannabis access and academic performance. *Review of Economic Studies*, 84(3):1210–1237.
- Mosher, C. and Akins, S. (2020). Recreational marijuana legalization in washington state: Benefits and harms. In Tom Decorte, G. P. and Bouchard, M., editors, *Legalizing Cannabis: Experiences, Lessons, and Scenarios*, pages 72–95. Routledge.
- National Conference of State Legislatures (2024). State medical cannabis laws.
- Nicholas, L. H. and Maclean, J. C. (2019). The effect of medical marijuana laws on the health and labor supply of older adults: Evidence from the health and retirement study. *Journal of Policy Analysis and Management*, 38(2):455–480.
- Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.
- Tyndall, J. (2021). Getting high and low prices: Marijuana dispensaries and home values. *Real Estate Economics*, 49(4):1093–1119.

A Appendix

A.1 Treatment and Control Group Definition

My sample consists of all postsecondary institutions in Washington State (approximately 127 campuses). I classify each institution as either “treated” or “control” based on its distance from the nearest dispensary. Specifically, the treatment group is defined as institutions located within a 10-minute driving time of at least one recreational dispensary, while the control group comprises institutions more than 10 minutes away. This 10-minute threshold provides a meaningful division: under this definition, about 92 institutions (roughly 75% of the sample) fall into the treatment group, and 35 institutions (25%) serve as controls in 2014. The number of institutions in the treatment category increases every year since more dispensaries opened. I choose 10 minutes as the cutoff because it yields a relatively balanced split between treated and control campuses, as shown in figure 6. By contrast, a very short threshold (e.g., 5 minutes) would classify only a small fraction of schools as treated, and a much longer threshold (e.g., 30 minutes) would effectively treat almost all institutions. Thus, the 10-minute criterion offers a reasonable balance for my analysis.

Figure 7 displays the geographic distribution of treated and control universities as of 2014. While colleges and universities tend to cluster in specific regions of Washington State, the assignment of nearby dispensaries appears quasi-random within these regions, reflecting the structure of the retail license lottery and its spatial implementation.

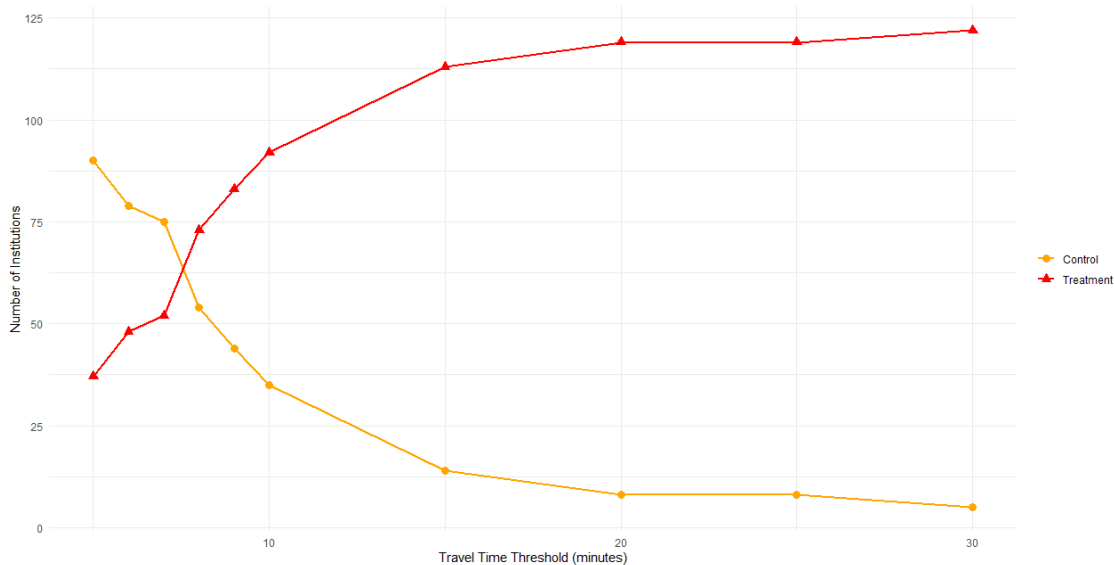


Figure 6: Number of Institutions by Threshold of Travel Time to Nearest Marijuana Dispensary

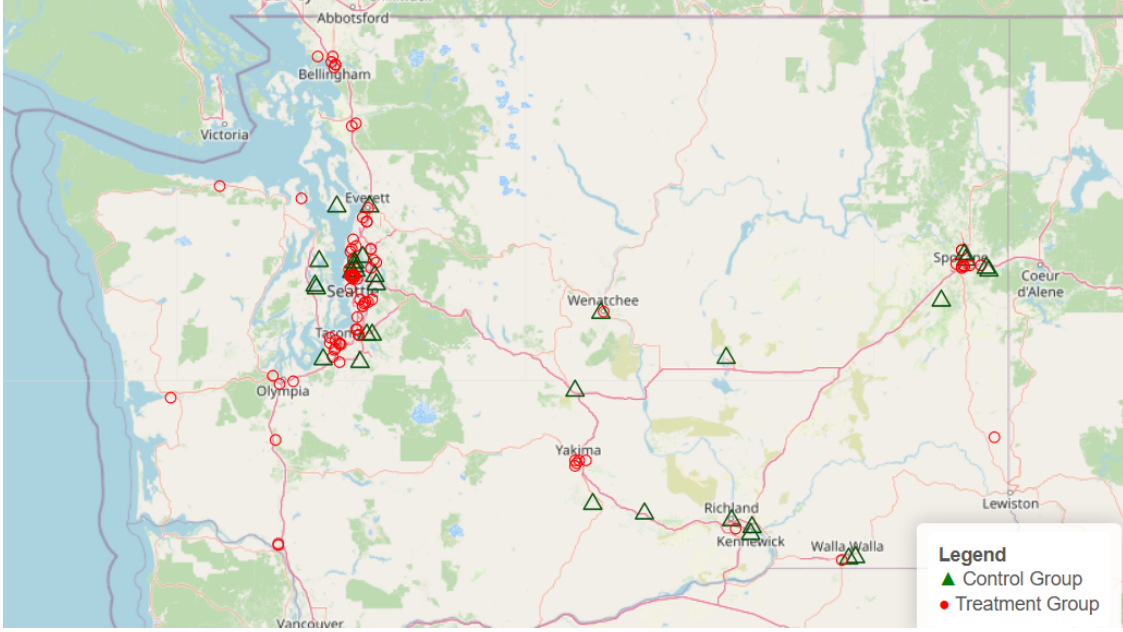


Figure 7: Treatment and Control Group Institutions in 2014

Overall Summary of ATT Based on Event-Study/Dynamic Aggregation			
ATT	Std. Error	[95% Conf. Int.]	
0.008	0.0285	[-0.0480, 0.0647]	
Signif. codes: ‘*’ confidence band does not cover 0			
Dynamic Effects			
Event time	Estimate	Std. Error	95% Simult. Conf. Band
-6	0.0124	0.0350	[-0.0761, 0.1009]
-5	-0.0095	0.0302	[-0.0940, 0.0750]
-4	0.0218	0.0320	[-0.0652, 0.1087]
-3	-0.0157	0.0307	[-0.1003, 0.0689]
-2	0.0081	0.0315	[-0.0782, 0.0945]
-1	0.0056	0.0274	[-0.0715, 0.0828]
0	0.0023	0.0241	[-0.0673, 0.0719]
1	-0.0187	0.0378	[-0.1124, 0.0750]
2	-0.0335	0.0455	[-0.1428, 0.0758]
3	-0.0289	0.0412	[-0.1285, 0.0707]
Estimation Details			
Control Group:	Not Yet Treated		
Anticipation Periods:	0		
Estimation Method:	Doubly Robust		

Table 4: Difference-in-Differences Analysis Results for Enrollment Rates

Distance Threshold (min)	ATT	Std. Error	95% Confidence Interval
5	0.0052	0.0261	[-0.0459, 0.0563]
6	-0.0127	0.0273	[-0.0661, 0.0407]
7	-0.0081	0.0280	[-0.0629, 0.0467]
8	0.0145	0.0255	[-0.0355, 0.0645]
9	0.0108	0.0279	[-0.0431, 0.0647]
10	-0.0075	0.0292	[-0.0648, 0.0498]
11	-0.0112	0.0304	[-0.0708, 0.0485]
12	0.0067	0.0287	[-0.0496, 0.0629]
13	0.0024	0.0302	[-0.0568, 0.0617]
14	-0.0150	0.0325	[-0.0788, 0.0488]
15	0.0041	0.0347	[-0.0643, 0.0725]

Table 5: Robustness Check: ATT Estimates for Enrollment Rates by Proximity Threshold